1. PHASE ALPHA: CRISIS SHELTERING ("OMG I NEED COVER NOW!!!")

■ **THE HOURLY EXPULSION TARGET:** Drop everything and establish a physical overhead barrier if rain begins, or if ambient temperatures drop below 50°F and wind speeds exceed 15 knots.

■ **THE 1-MINUTE PLYWOOD OR TARP SHEAR:** Locate a 10x10 waterproof tarp, heavy polyethylene sheet, or structural plywood board. Secure a primary anchor line around a solid standing tree trunk exactly 4 feet above ground level.

■ **THE EMERGENCY PLOW-POINT PITCH:** Secure one corner of the tarp to the tree trunk anchor. Pull the opposite diagonal corner tight away from the tree and anchor it directly into the ground with a heavy-duty stake at a 45-degree angle pointing away from the line. Stake the remaining two side corners flat to the dirt. This forms a low-profile, aerodynamic wedge that deflects high winds and sheds heavy downpours over your back.

■ **THE CONVECTIVE EXCLUSION ZONE:** Never sit or lie down directly on bare, cold, damp dirt or concrete floors. The cold ground will rapidly pull thermal energy out of your body via conductive transfer. Pile dry leaves, pine straw, or tree branches into a dense, 6-inch thick mattress inside your shelter to form an insulated air gap.

SITE SELECTION CONSTRAINTS MATRIX

CRITICAL PARAMETER	MANDATED FIELD CONFIGURATION RULE	MECHANICAL HAZARD AVOIDANCE
HIGH GROUND	Position shelter a minimum of 50 vertical feet above visible valley beds, dry creek runs, or drainage channels.	Completely eliminates flash-flood risks, mud slides, and dense cold-air pooling pockets.
OVERHEAD AUDIT	Inspect the immediate canopy for dead tree limbs ('widowmakers') or loose, unstable structural masonry blocks.	Prevents severe crush injuries, skull fractures, or sudden shelter collapse from wind drops.
THERMAL SHIELD	Pitch shelter footprint downward/lee-side of dense foliage stands, thick hedgerows, or rock walls.	Minimizes convective wind-chill vectors, reducing your internal heating fuel consumption.

2. PHASE BRAVO: LONG-TERM SUSTAINABILITY & STRUCTURAL ENGINEERING

■ **THE DEAD-FALL LEAN-TO FRAMING MATH:** Select a solid, living tree or rock ledge to anchor a single, heavy, 8-foot long ridgepole log (4-inch diameter). Prop the log horizontally at a 45-degree lean angle.

■ **TRIANGULAR SUPPORT STRUCTURING:** Lean thick support branches (2-inch diameter) along one side of the ridgepole log spaced exactly 6 inches apart. Ensure each branch forms a rigid 60-degree angle with the ground surface. This exact slope prevents collapse under heavy rain loads and optimizes rainwater shedding.

■ **THE THERMAL R-VALUE INSULATION BLANKETING:** Layer green boughs, thick pine needles, or packed dry leaves over the support frame. Build this insulation layer until it reaches a physical thickness of 12 inches. A 12-inch layer of dead leaves creates a dense, dead-air pocket matrix providing an estimated R-value insulation equivalent of R-4 to R-6, trapping internal body warmth and fire radiation cleanly.

■ **GROUND PALLET CRADLING:** Lay down structural wooden pallets, dry foam insulation pads, or parallel dead hardwood logs to build an elevated sleeping platform inside the core footprint. Cover this platform with a waterproof ground cloth to keep rising ground moisture out of your bedding materials.

3. HOURLY STRUCTURAL INTEGRITY MAINTENANCE LOG

RECURRING HOUR LOG	ANCHOR STAKE NODES	ROOF INTEGRITY MODES	ACTION EXECUTED (■)
HOUR _____	[ ] Rigid [ ] Loose	[ ] Shedding [ ] Pooling	[ ] Tensioned Hitches [ ] Cleared Debris
HOUR _____	[ ] Rigid [ ] Loose	[ ] Shedding [ ] Pooling	[ ] Tensioned Hitches [ ] Cleared Debris
HOUR _____	[ ] Rigid [ ] Loose	[ ] Shedding [ ] Pooling	[ ] Tensioned Hitches [ ] Cleared Debris

STRUCTURAL ENGINEERING & COLD COMFORT PLANNING PANEL
AUTHORITATIVE REFERENCES: Federal Emergency Management Agency (FEMA) Ready.gov Relocation Guidelines, US Army Survival Manual FM 21-76 Structural Frameworks.

LEGAL DISCLAIMER: Emergency shelter designs are independent aids for short-term preservation. They do not substitute for code-compliant permanent structural dwellings. Never use open-flame heaters inside unventilated, enclosed plastic layouts.